Cluster-Adaptive Prompting for Multi-Turn Travel and Tourism Conversational Systems

Interim Report

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GitHub repository: https://github.com/surnella/QM640-Dialogue-Clustering-Capstone

Contents

Cluster-Adaptive Prompting for Multi-Turn Travel and Tourism Conversational Systems 4

Background and Context.....4

Problem Statement.....4

Purpose of the Study.....5

Interim Project Status.....5

Scope and Objectives.....6

Inclusions.....6

Exclusions.....6

Feasibility.....6

Research Questions and Hypotheses.....7

Literature Survey.....7

Literature Review Approach.....7

Thematic Review and Research Gaps.....8

Data Description.....9

Data Sources and Access.....9

Dataset Overview.....10

Data Dictionary.....10

GitHub Data Availability Statement.....11

Dataset Limitations.....12

Analysis.....12

Data Cleaning.....12

Exploratory Data Analysis Results 13
Modeling 15
Research Design and Sample Size 15
Choice of Models With Justification 17
Features Included and Feature Engineering 21
Evaluation Metrics 22
Preliminary Results 23
RQ1: Clustering Algorithms and Primary Archetypes (Complete) 23
RQ1 Sensitivity Analysis: Refined Archetypes (Complete) 25
RQ2: Cluster-Adaptive Few-Shot Prompting (Complete) 28
RQ3: Rules Versus Prompting (Groundwork Complete) 30
RQ4: Turn Economy Across Archetypes (Groundwork Complete) 30
Interim Limitations and Risks 31
Next Steps 31
References 32
Appendix A 37
Appendix B 39
Appendix C 40
Appendix D 41
Appendix E 44

Cluster-Adaptive Prompting for Multi-Turn Travel and Tourism Conversational Systems

Background and Context

Conversational artificial intelligence (AI) agents are now widely used in travel and tourism to support destination discovery, itinerary planning, and booking (Jannach et al., 2021). Travelers differ in how they converse: some state every constraint in the first message, some revise budgets and dates repeatedly, and others rely on the agent to lead the search (Inaba et al., 2022). Most travel assistants, however, use one static prompt for every user, which produces repetitive clarification questions, mismatched pacing, and abandoned sessions.

Designers usually adapt a large language model (LLM) agent's style through prompt engineering, for example, by inserting example dialogues into the prompt. Prompt-only adaptation creates two risks (Rebedea et al., 2023): the model can copy stale details such as earlier prices, dates, or destinations from the examples, and style instructions compete with business constraints, so the agent may propose an itinerary that violates passport, visa, capacity, or discount policies. This study therefore separates the two concerns: conversation archetypes learned from historical dialogues select in-context examples that steer style, and a declarative rule engine outside the prompt checks policy.

Problem Statement

Travel conversational agents lack a data-driven way to recognize a user's conversation archetype and adapt their style to it, and prompt-based style steering can introduce factual leakage and policy violations. The study measures how four independent variables—clustering algorithm (RQ1), in-context conditioning strategy (RQ2), rule-engine mode (RQ3), and conditioning strategy crossed with user archetype (RQ4)—affect cluster separation, style convergence, slot accuracy, exemplar leakage, rule adherence, and turn economy.

Purpose of the Study

The purpose of this quantitative study is to

1. discover and interpret conversation archetypes in human travel dialogues and identify the clustering algorithm that separates them best (RQ1);
2. measure whether archetype-matched in-context examples move an LLM agent's responses toward the target archetype's style more than zero-shot or static examples do (RQ2);
3. isolate the effects of a declarative rule engine and of cluster-adaptive prompting on slot accuracy, exemplar leakage, and rule adherence (RQ3); and
4. estimate whether cluster-adaptive prompting changes turn economy differently across archetypes (RQ4).

Interim Project Status

- *Completed:* data ingestion, quality audit, cleaning, exploratory analysis, and power analysis; RQ1 end to end, including a sensitivity analysis that produced three refined archetypes (Decisive Booker, Open Explorer, and Rigid-Constraint Seeker); and the RQ2 experiment (99 LLM generations).
- *In progress:* groundwork for RQ3 and RQ4—the rule engine is verified, the corpus is audited for policy information, and a human baseline for turn economy is measured.
- *Pending:* repeating RQ2 with the refined archetypes, the RQ3 2 × 2 experiment, the RQ4 simulation, robustness checks, and the final report (Appendix B).

Scope and Objectives

Inclusions

The study includes English-language, multi-turn, human–human travel dialogues from two public corpora: all 1,369 Microsoft Frames dialogues and a 1,500-dialogue sample of MultiWOZ 2.2 dialogues that involve at least one travel service (hotel, train, taxi, or attraction), each with at least three turns. The pre-specified analyses use one embedding model (Sentence-BERT all-MiniLM-L6-v2), three clustering algorithms, and one LLM family (Gemini Flash); the RQ1 sensitivity analysis adds Gemini embeddings and LLM ratings of user behavior. Scope is contained by using Frames as the only corpus for learning, retrieval, and experiments, and by having every research question reuse one pipeline, one set of archetype assignments, and one held-out partition, so that each later question adds an experimental factor rather than a new method.

Exclusions

No claim is made that Frames archetypes transfer to MultiWOZ or other domains; MultiWOZ serves only as a descriptive projection check. Proprietary, enterprise, or customer data, new human-subject data, and synthetic training dialogues are excluded; the only synthetic elements are the author-defined policy rules and scenarios used in RQ3. Non-English, voice, restaurant-only, and sub-three-turn dialogues, model fine-tuning, other LLM providers, live deployment, real bookings, and user-satisfaction outcomes are also out of scope.

Feasibility

Both corpora are freely downloadable, and the core pipeline runs on a laptop in under 15 minutes. RQ2 required 99 LLM calls, the RQ1 sensitivity analysis 4,860 annotation calls, and

RQ3 and RQ4 together need fewer than 2,000, all within the free tier of the Gemini application programming interface. Every response is cached so that interrupted runs resume.

Research Questions and Hypotheses

Table 1 states each research question with its hypotheses and the analysis that tests it.

Table 1

Research Questions, Hypotheses, and Planned Analyses

RQ	Research question	Hypotheses	Analysis
RQ1	Which clustering algorithm (K-Means, agglomerative, HDBSCAN) gives the best separation of dialogue-trajectory embeddings, and which archetypes emerge?	H_0: mean Silhouette is equal across algorithms. H_1: at least one mean differs.	One-way ANOVA (Welch and Kruskal–Wallis as robustness checks); Tukey and Games–Howell post hoc
RQ2	Does cluster-adaptive few-shot prompting increase style convergence relative to zero-shot and static few-shot prompting?	H_0: mean style cosine is equal across conditions. H_1: cluster-adaptive > static and zero-shot.	One-way repeated-measures ANOVA; Greenhouse–Geisser correction; Holm-adjusted paired contrasts
RQ3	Do a declarative rule engine and cluster-adaptive prompting affect slot F1, exemplar leakage, and rule adherence?	H_0: no main or interaction effects. H_1: rules raise adherence and reduce leakage without lowering style convergence.	2 × 2 between-subjects factorial ANOVA
RQ4	Does cluster-adaptive prompting change turns to goal completion and redundant elicitation differently across archetypes?	H_0: no conditioning × archetype interaction. H_1: the reduction in turns differs by archetype.	2 × 3 factorial ANOVA on log turns

Note. RQ = research question; ANOVA = analysis of variance. Significance level $\alpha = .05$ for all tests.

Literature Survey

Literature Review Approach

Sources were located in the ACL Anthology, Google Scholar, and arXiv with combinations of the terms task-oriented dialogue corpus, dialogue clustering, sentence embeddings, in-context example selection, LLM guardrails, policy compliance, and LLM annotation. Peer-reviewed papers and widely cited preprints were retained when they supplied a

corpus, method, or measure used in this study, and seminal statistical sources were added for each metric and test. Each source was mapped to the research questions it informs; the relevance matrix of the closest studies is in Appendix C.

Thematic Review and Research Gaps

Task-oriented dialogue corpora. MultiWOZ (Budzianowski et al., 2018) is the standard multi-domain benchmark, and MultiWOZ 2.2 (Zang et al., 2020) corrected its state annotations, but it covers local services in one city and contains no flights or package negotiation. Microsoft Frames (El Asri et al., 2017) was designed for exactly that setting: users negotiate flight-and-hotel packages under a budget and compare several offers before deciding. Frames is therefore the primary corpus, and MultiWOZ 2.2 is an external comparison.

Dialogue representation and clustering. Sentence-BERT (Reimers & Gurevych, 2019) yields embeddings whose cosine distances reflect semantic similarity, enabling whole-dialogue representations without task-specific training. K-Means, Ward agglomerative clustering, and HDBSCAN (McInnes et al., 2017) assume different cluster shapes and should be compared rather than chosen a priori; quality is judged with the Silhouette coefficient (Rousseeuw, 1987) and the Davies–Bouldin index (Davies & Bouldin, 1979), and stability with the adjusted Rand index under resampling (Hennig, 2007; Hubert & Arabie, 1985). UMAP (McInnes et al., 2018) and scikit-learn (Pedregosa et al., 2011) support inspection and reproducibility.

In-context exemplar selection. LLMs can perform new tasks from a few in-context examples (Brown et al., 2020), and examples retrieved by embedding similarity outperform random ones (Liu et al., 2022); FAISS (Johnson et al., 2021) makes such retrieval fast. These studies evaluate single-turn accuracy, not conversational style, and retrieve from one global pool

rather than from behaviorally defined sub-pools. LLMs can also annotate text as reliably as trained crowd workers (Gilardi et al., 2023), which supports LLM ratings of user behavior.

Policy compliance and evaluation. NeMo Guardrails (Rebedea et al., 2023) moves policy logic into programmable rails outside the model, τ-bench (Yao et al., 2024) shows that policy adherence must be measured separately from task success, and TD-EVAL (Acikgoz et al., 2025) argues that dialogue-level similarity such as BERTScore (Zhang et al., 2020) must be complemented by turn-level checks of factual consistency and policy compliance.

Three gaps follow. No reviewed study derives conversation archetypes from dialogue trajectories and uses them to select in-context examples; studies that steer style do not measure whether examples leak stale facts, and studies that enforce policy do not adapt style; and the turn-economy effect of style adaptation across user archetypes has not been tested. RQ1 and RQ2 address the first gap, RQ3 the second, and RQ4 the third.

Data Description

Data Sources and Access

Microsoft Frames (El Asri et al., 2017) contains 1,369 English Wizard-of-Oz dialogues in which a user role-player negotiates a flight-and-hotel package with a wizard who searches a database of fictional cities; it is available from Microsoft Research at https://www.microsoft.com/en-us/research/project/frames-dataset/. MultiWOZ 2.2 (Zang et al., 2020) contains 10,438 dialogues with per-turn dialogue states and is available at https://github.com/budzianowski/multiwoz. Both corpora were downloaded in full and are public, were collected with consent for research use, and identify role-players only by codes, so no new human data are collected. Only public dialogue text with fictional cities is sent to the

LLM, and archetypes are described only by conversational behavior, never by inferred demographics.

Dataset Overview

Table 2 summarizes the records, variables, time coverage, unit of analysis, and outcome variables.

Table 2

Dataset Overview

Attribute	Microsoft Frames (primary)	MultiWOZ 2.2 (comparison)
Records	1,369 dialogues; 1,334 analytic (golden partition 933, held-out partition 401)	10,438 dialogues; seeded sample of 1,500; 1,499 analytic
Variables used	Utterance text, speaker, dialogue acts with slot–value arguments, task-success label, user rating, role-player codes (Table E1)	Utterance text, speaker, services, per-turn dialogue state (Table E1)
Time period	Released 2017; timestamps record collection time only and are not analyzed	Released 2018 (version 2.2 in 2020); no timestamps
Unit of analysis	Dialogue; clustering run (RQ1 algorithm test); generated agent turn (RQ2)	Dialogue (descriptive projection only)
Target or outcome	No supervised target; outcomes are Silhouette (RQ1), style convergence (RQ2), slot F1, leakage, and adherence (RQ3), turns and redundant elicitation (RQ4)	Similarity to the nearest Frames centroid

Data Dictionary

Table 3 defines the analysis variables. The raw source fields read from each corpus and the canonical schema produced by preprocessing are listed in Tables E1 and E2 in Appendix E.

Table 3

Data Dictionary of the Analysis Variables

Variable	Role	Type	Operational definition	RQ
Clustering algorithm	IV	Nominal (3)	K-Means, Ward agglomerative, HDBSCAN	1
Silhouette coefficient	DV	Continuous [−1, 1]	Cosine Silhouette of a clustering run	1

Variable	Role	Type	Operational definition	RQ
Davies–Bouldin index	DV	Continuous ≥ 0	Mean cluster similarity ratio (lower is better)	1
Archetype (primary)	IV / moderator	Nominal (3)	Nearest frozen *K*-Means centroid (cosine, MiniLM trajectory vectors)	1, 2
Archetype (refined)	IV / moderator	Nominal (3)	*K*-Means on the Gemini embedding and LLM behavior profile of the first nine user messages	1, 3, 4
Behavior rating	Feature	Ordinal (1–5)	Ten LLM ratings of user behavior (Appendix D)	1
Conditioning strategy	IV	Nominal (3)	Zero-shot, static few-shot, cluster-adaptive few-shot ($k = 3$ exemplars)	2–4
Style convergence	DV	Continuous [−1, 1]	Cosine between response and target archetype's agent-style centroid	2, 3
Gold similarity	DV	Continuous [−1, 1]	Cosine between response and the human reply	2
Rule-engine mode	IV	Nominal (2)	Active or off	3
Slot F1	DV	Proportion	F1 of slot values in the response against the dialogue state	3
Exemplar leakage rate	DV	Proportion	Entities in the response that occur in exemplars but not in the dialogue / all entities	3
Rule adherence rate	DV	Proportion	1 − (violating agent turns / agent turns)	3
Turns to goal completion	DV	Count (log)	Turns until all required slots are filled and an offer is accepted	4
Redundant elicitation rate	DV	Proportion	Agent requests for an already-informed slot / agent turns	4

Note. IV = independent variable; DV = dependent variable. All cosine measures use de-lexicalized text and all-MiniLM-L6-v2 embeddings.

GitHub Data Availability Statement

Code, configuration, processed data, and results are in the public repository (https://github.com/surnella/QM640-Dialogue-Clustering-Capstone). The raw corpora are not redistributed: data/raw/ is excluded from version control and is recreated by downloading Frames and MultiWOZ 2.2 from the sources above. Processed data (cleaned samples, partitions, archetype labels, LLM annotations, and saved prompts) are in data/interim/, result tables and figures in outputs/interim/, and the analysis code in src/interim_pipeline/. This report is generated from those files, so rerunning the steps in Appendix A reproduces every number.

Dataset Limitations

Frames dialogues were produced by 11 user role-players and 12 wizards following written scenarios, so archetypes may partly reflect individual habits, and its cities and prices are fictional. MultiWOZ covers one city without flights or package negotiation and records price only as a category. Several slots are often absent (Table E3): Frames mentions children in only 18.4% of dialogues and trip duration in 61.8%, MultiWOZ never records a return date, and neither corpus contains visa, passport, or discount information, so RQ3 must inject these conditions. Neither corpus records demographics or real booking outcomes; the Frames task-success label is wizard-reported and imbalanced (94.2% successful), and 3 dialogues lack a user rating.

Analysis

Data Cleaning

Table 4 is the cleaning log. Missing slot values are not imputed, because a slot that is never mentioned is correctly unobserved for a dialogue-state tracker, and turn-count outliers are retained because long negotiations are genuine behavior of interest.

Both corpora were parsed into one canonical format. Frames labels such as *dst_city* and MultiWOZ labels such as *train-destination* were mapped to a two-tier schema: Tier 1 holds nine universal travel slots (origin, destination, departure and return dates, duration, adults, children, budget, and accommodation tier), and Tier 2 holds policy slots (visa processing window, passport validity, package type, room sharing, and discount) used only by the rule engine. After cleaning, the 1,334 analytic Frames dialogues were split once (seed 42, stratified on task success) into a golden partition (D_{gold}, 70%, $n = 933$) used to fit clusters and build the exemplar pool, and a held-out partition (D_{test}, 30%, $n = 401$) used only for evaluation.

Table 4

Data Cleaning Log

Issue	Variables	Detection	Frames	MultiWOZ	Treatment and rationale
Duplicate identifiers	Dialogue ID	Uniqueness check	0	0	None needed
Duplicate full text	All utterances	Hash of the dialogue text	0	1	Removed; would double-weight one dialogue
Empty utterances	Utterance text	Length check	0	0	None needed
Fewer than three turns	Turns	Turn count	35	0	Excluded; no negotiation trajectory
Turn-count outliers	Turns	> Q3 + 1.5 × IQR	22 (> 32.5)	13 (> 27.0)	Retained; decay weighting and log turns limit influence
Missing slot values	Tier 1 and 2 slots	Slot never mentioned	See Figure 2	See Figure 2	Not imputed; unobserved is the correct state
Missing user rating	User rating	Null check	3	Not available	Retained; rating is descriptive only
Corpus-specific labels	Slots, speakers	Schema mapping	All	All	Mapped to one two-tier schema; speakers become USER or AGENT
Entity values in text	Utterance text	Slot values and numbers	All	All	Replaced with slot tokens so examples carry style, not facts
Analytic sample			1,334	1,499	

Note. IQR = interquartile range. Restaurant-only MultiWOZ dialogues were excluded before sampling. No dialogue lacked a task-success label (Frames).

Exploratory Data Analysis Results

Table 5 describes the analytic samples. Frames and MultiWOZ dialogues have similar mean lengths (M = 14.93 and 15.17 turns), but Frames lengths vary more (SD = 6.74 vs. 4.46) and have a longer right tail (Figure 1). The golden and held-out partitions are nearly identical in length, utterance length, and success rate, which confirms that the stratified split worked as intended.

Table 5

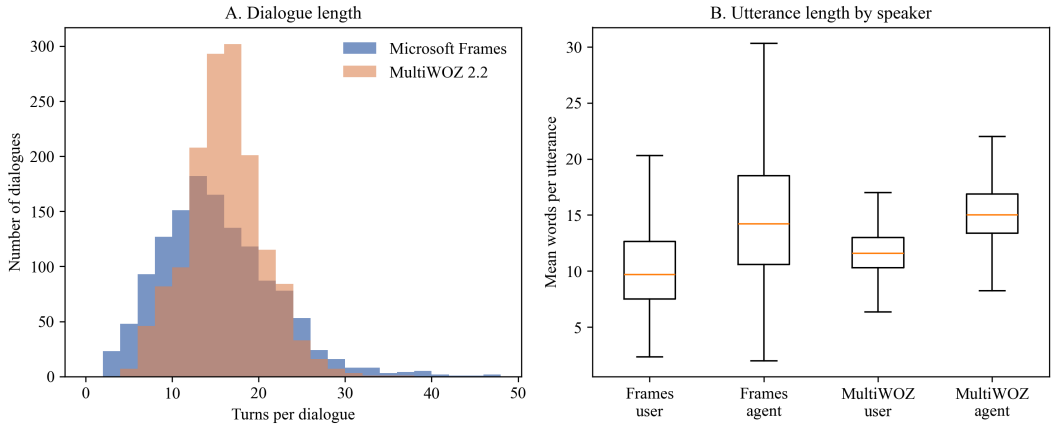
Descriptive Statistics of the Analytic Samples

Sample	*N*	Turns *M* (*SD*)	*Mdn*	Range	User words	Agent words	Task success (%)
Microsoft Frames	1,334	14.93 (6.74)	14	3–48	10.6	15.0	94.2
MultiWOZ 2.2	1,499	15.17 (4.46)	16	4–36	11.7	15.1	—
Frames golden partition	933	14.95 (6.75)	14	3–48	10.7	15.3	94.2
Frames held-out partition	401	14.90 (6.71)	14	3–43	10.4	14.5	94.3

Note. User and agent words are mean words per utterance. Task success is the wizard-reported label (not available in MultiWOZ).

Figure 1

Distribution of Dialogue Length by Corpus



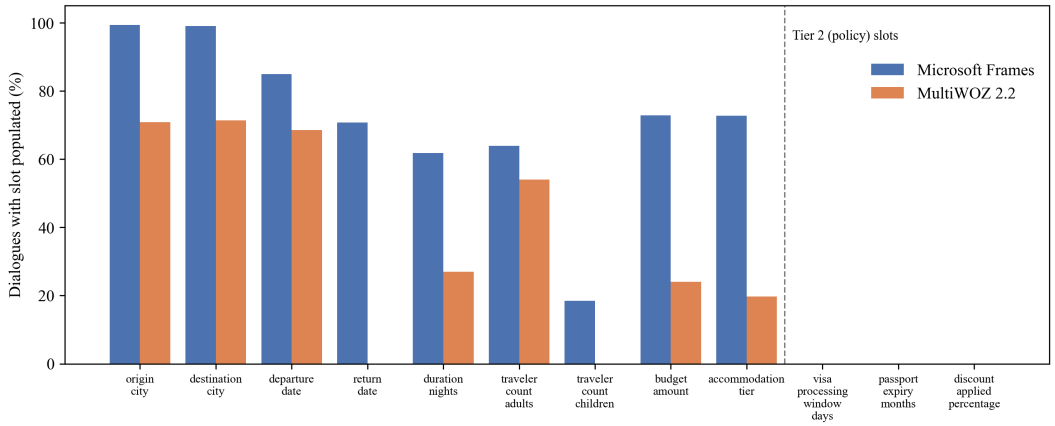
Note. Turn counts after cleaning. Frames $N = 1{,}334$; MultiWOZ $N = 1{,}499$.

Figure 2 compares slot completeness (exact percentages in Table E3). Frames users state origin and destination in almost every dialogue and a budget in most, whereas MultiWOZ dialogues mention travel slots far less often and never mention return dates. Three insights shape the analysis. First, Frames is the only corpus with enough budget and package information for the negotiation behavior that the archetypes are meant to capture. Second, length differences between the corpora mean that any archetype tied to dialogue length will be distributed

differently across corpora, which motivates treating MultiWOZ only as a descriptive check. Third, the absence of policy slots in both corpora confirms that RQ3 requires injected scenarios.

Figure 2

Slot Completeness by Corpus



Note. Percentage of dialogues that mention each canonical slot at least once.

Modeling

Research Design and Sample Size

The study uses a quantitative design with one observational component (RQ1) and three controlled experiments (RQ2 to RQ4). All random steps use seed 42, and leakage is prevented by keeping the partitions disjoint, freezing centroids before any held-out dialogue is seen, routing and retrieving with only the turns available at generation time, and using the human reply only for scoring, never in a prompt.

Each sample size was computed for the exact test used later, with a medium effect $f = .25$ (Cohen, 1988), $\alpha = .05$, and power .80, because no prior study reports effect sizes for these comparisons. Power was computed from the noncentral F distribution and checked against G*Power 3 (Faul et al., 2007):

$$1 - \beta = 1 - F_{nc}(F_{crit}; df_1, df_2, \lambda), \quad F_{crit} = F_{1-\alpha}(df_1, df_2) \quad (1)$$

where F_{nc} is the cumulative noncentral F distribution and $\lambda = f^2 N$ for between-subjects designs. The smallest N with power ≥ .80 (N_{min}) was rounded up to a balanced design and, for LLM experiments, increased by 10% for failed generations. For RQ1, for example, $N = 158$ gives $\lambda = .0625 \times 158 = 9.875$, $F_{crit}(2, 155) = 3.054$, and power = .802, so 159 clustering runs (53 per algorithm) are used. Table 6 gives the computation for every research question. The experiments draw from the held-out partition, so the binding requirement is the largest experimental sample, 180 dialogues, which is smaller than $|D_{test}| = 401$.

Table 6

Sample Size Calculations Matched to Each Planned Analysis

RQ	Planned analysis	df at N_{min}	λ	F_{crit}	N_{min}	Power	N_{final} (per group)	Final power
RQ1	One-way ANOVA (3 algorithms)	2, 155	9.88	3.05	158	.802	159 (53)	.805
RQ2	One-way repeated-measures ANOVA (3 conditions)	2, 54	10.50	3.17	28	.812	33 (11)	.878
RQ3	2 × 2 factorial ANOVA	1, 124	8.00	3.92	128	.801	144 (36)	.846
RQ4	2 × 3 factorial ANOVA	2, 152	9.88	3.06	158	.802	180 (30)	.855

Note. All calculations use $f = .25$ and $\alpha = .05$. RQ1 units are clustering runs on subsamples of the golden partition; RQ3 and RQ4 units are test dialogues. RQ2 units are test dialogues measured under $m = 3$ conditions, with $\lambda = f^2 N m \varepsilon / (1 - \rho)$, $\rho = .50$, and $\varepsilon = 1$ (Faul et al., 2007). RQ2 to RQ4 include a 10% buffer for failed generations.

Choice of Models With Justification

No predictive model is trained. Each research question is answered with an unsupervised or experimental model and a statistical test matched to its design; baselines and assumptions are stated for each.

RQ1: Clustering Algorithms and Archetype Definition

K-Means ($k = 3$, 10 initializations), Ward agglomerative clustering ($k = 3$), and HDBSCAN, which chooses the number of clusters itself, were each applied to 53 independent random subsamples of 500 golden dialogues and compared with a one-way ANOVA on Silhouette. HDBSCAN noise points were assigned to the nearest centroid so that every dialogue receives an archetype. The baselines are a permutation baseline (K-Means labels shuffled within each subsample) and K-Means as the conventional reference.

Independence of runs. Each run uses its own subsample; because subsamples share dialogues, the test compares algorithms on this corpus rather than across datasets.

Normality. Shapiro–Wilk tests did not reject normality (K-Means $W = .965$, $p = .120$; agglomerative $W = .979$, $p = .477$; HDBSCAN $W = .980$, $p = .524$).

Homogeneity of variance. Levene's test was significant, $W = 12.75$, $p < .001$, so Welch's ANOVA, Games–Howell comparisons, and the Kruskal–Wallis test are reported as well.

Meaningful embedding geometry. Cosine distance is assumed to reflect conversational behavior; this was checked with behavioral features not used in clustering (Table 10).

Cross-corpus invariance. The synopsis assumed that Frames archetypes would apply to MultiWOZ; projection onto the frozen centroids did not support this.

A sensitivity analysis varied whose turns are represented (whole dialogue or user turns), how (MiniLM or Gemini embeddings, with or without an LLM behavior profile), and how much

of the conversation (first 3, 5, 7, or 9 user messages, or all), under a selection rule fixed before results were inspected. Stability was the mean adjusted Rand index (ARI) between K-Means fits on two random 80% subsamples, repeated 50 times (Hennig, 2007). Eligible solutions had three to eight clusters, ARI ≥ .60, and a smallest cluster ≥ 5%; for each length, the eligible solution that a router could predict best from the first five turns (Cohen's κ) was retained, and the most stable retained solution was adopted. Two-cluster solutions were excluded in advance as too coarse for archetype-specific prompting. The confirmatory RQ1 test is unchanged; this analysis only selects the archetype labels used later.

Valid behavior ratings. LLM ratings are assumed to measure user behavior; this was tested by re-annotation and by correlation with behaviors counted from the corpus annotations (see Features).

RQ2: Repeated-Measures Prompting Experiment

The experiment sampled 33 held-out dialogues, 11 per primary archetype (it was run before the sensitivity analysis). The first three turns form the prefix, and Gemini 2.5 Flash (temperature 0.2, reasoning disabled) generated the next agent turn under three conditions: zero-shot, static few-shot (the same three random golden exemplars for every dialogue), and cluster-adaptive few-shot, in which the prefix is routed to the nearest archetype centroid and the three most similar golden dialogues are retrieved from that archetype's FAISS index. Zero-shot and static few-shot are the baselines, and unclustered global nearest-neighbor retrieval (Liu et al., 2022) was compared in the pilot. A one-way repeated-measures ANOVA with Holm-adjusted contrasts tests the conditions; all 99 generations completed ($N = 33$).

Sphericity. Mauchly's test was not significant for style convergence, $W = .95$, $\chi^2(2) = 1.59$, $p = .451$, or gold similarity, $W = .97$, $p = .610$; Greenhouse–Geisser-corrected results (Greenhouse & Geisser, 1959) are reported for consistency.

Normality of difference scores. Shapiro–Wilk p values for the paired differences ranged from .003 to .616, so the Friedman test is reported as a distribution-free check.

Valid style metric and correct routing. The style score assumes that a turn written for an archetype is closer to its agent-style centroid, and routing assumes that the prefix identifies the archetype; the pilot tested both.

Exemplars carry style, not facts. De-lexicalization is assumed to remove factual content; this is tested as exemplar leakage in RQ3.

RQ3: Rules-by-Prompting Factorial Experiment

RQ3 randomly assigns 144 held-out dialogues to a 2 × 2 between-subjects design (rule engine active or off × zero-shot or cluster-adaptive prompting with the refined archetypes). Each dialogue state is augmented with seeded synthetic Tier 2 values, half of which violate at least one rule. When active, the engine evaluates the state before generation and adds violations to the prompt as required actions. The baseline cell is zero-shot prompting with rules off. Table 7 lists the five author-defined rules, which do not represent any company's policy.

Deterministic rule evaluation. All 14 of 14 boundary tests passed, including values exactly at each threshold and the domestic and group-package exemptions.

Decoupling. Rules are assumed to act independently of prompt style; this is the interaction term, so it is tested rather than assumed.

Residual normality and equal variances. Shapiro–Wilk and Levene tests will be applied; because adherence proportions are likely near 0 or 1, an aligned rank transform ANOVA or a logistic model will be used if they fail.

International travel. Frames cities are fictional, so all trips are treated as international for the visa, passport, and budget rules.

Table 7

Declarative Policy Rules Used in RQ3

Rule	Condition	Action	Severity
Visa Processing Lead Time	departure_lead_days < 15; is_international = true	block and flag	fatal
Passport Validity Threshold	passport_expiry_months < 6; is_international = true	block and flag	fatal
Maximum Adult Capacity per Single Booking	traveler_count_adults > 8; package_type = 'FIT'	convert to group	warning
Maximum Discretionary Sales Discount	discount_applied_percentage > 15.0	clamp discount	fatal
Minimum Feasible Per-Pax International Budget	budget stated; budget_amount < 500; is_international = true	prompt realistic budget	warning

Note. Rules are stored in a YAML file and evaluated outside the LLM by src/rule_engine.py.

RQ4: Turn-Economy Simulation

RQ4 uses 180 held-out dialogues, 30 per cell of a 2 × 3 design (zero-shot or cluster-adaptive prompting × the three refined archetypes; the held-out partition contains 175, 99, and 127 dialogues of these archetypes). An LLM user simulator receives the dialogue's user goal and its archetype profile (Table 12) and converses with the agent until the goal is met or 30 turns elapse. The zero-shot agent is the baseline, and the human dialogues give a reference level. A 2 × 3 factorial ANOVA on log turns tests the interaction.

Log-normal turn counts. Turn counts are right-skewed and are analyzed on the log scale; residuals will be checked with Shapiro–Wilk tests.

Equal variances. In the human baseline, Levene's test on log turns was significant, $W = 38.01$, $p < .001$, so heteroscedasticity-robust (HC3) standard errors will be used.

Comparable goals. Turn counts are compared only within Frames, so the schema need not be invariant across corpora.

Archetype effect by construction. Archetypes partly reflect dialogue length, so an archetype main effect is expected and is not a finding; the tests of interest are the conditioning effect and the interaction.

Valid persona simulation. Simulated user turns will be annotated with the same codebook and tested for assignment to the intended archetype above chance.

Features Included and Feature Engineering

Trajectory vectors (primary archetypes, RQ2 routing). Each de-lexicalized turn t of a dialogue with T turns was embedded with Sentence-BERT as a 384-dimensional vector $\mathbf{e}_t$, and the dialogue vector is the decay-weighted mean

$$\mathbf{v} = \Sigma_t \, w_t \mathbf{e}_t, \quad w_t = \exp[-\lambda_d(T - t)] \,/\, \Sigma_j \exp[-\lambda_d(T - j)] \quad (2)$$

with $\lambda_d = 0.15$ and L2 normalization, which emphasizes how a dialogue develops and closes. Structural features (turns, revisions, hesitations, offers) are excluded from the vector and used only to interpret clusters.

Hybrid features (refined archetypes). Two equally weighted blocks were combined: the first 20 principal components of the Gemini embedding (gemini-embedding-001, 3,072 dimensions; Lee et al., 2025) of the user's messages, and an LLM behavior profile. For the profile, Gemini 2.5 Flash (temperature 0, output constrained to a JSON schema) rated each user against a 16-field codebook (Table D1): ten 1–5 ratings such as price sensitivity, destination flexibility, decisiveness, and revision tendency, plus goal and tone indicators; outcome and party

type were annotated but not used. Both blocks were computed on the same first *M* user messages. Re-annotating 100 random dialogues at temperature 0.7 gave quadratic-weighted κ from .87 to .96 for the ratings, and the ratings correlated with behaviors counted from the Frames acts, for example revision tendency with slot revisions (ρ = .73) and question asking with request rate (ρ = .65; Table D2).

Exemplar features (RQ2 to RQ4). Retrieval keys embed the same number of opening turns as the test prefix, so neither side uses future turns; exemplars are de-lexicalized and truncated to six turns.

Evaluation Metrics

Table 8 defines every outcome metric with its formula.

Table 8

Evaluation Metrics and Formulae

Metric	Formula	RQ	Interpretation
Silhouette	$s(i) = [b(i) - a(i)] / \max\{a(i), b(i)\}$, averaged over i; a, b = mean cosine distance to own and nearest other cluster	1	−1 to 1; higher is better separated
Davies–Bouldin index	$DB = (1/k)\ \Sigma_i \max_{j \neq i} (\sigma_i + \sigma_j) / d(\mathbf{c}_i, \mathbf{c}_j)$	1	≥ 0; lower is better
Adjusted Rand index	$ARI = (RI - \mathrm{E}[RI]) / (\max RI - \mathrm{E}[RI])$	1	0 = chance agreement; 1 = identical partitions
Cohen's κ	$\kappa = (p_o - p_e) / (1 - p_e)$	1	Routing and rating agreement beyond chance
Cramér's V	$V = \sqrt{[\chi^2 / (N(\min(r, c) - 1))]}$	1	0 to 1; association of archetype with a category
Style convergence	$\cos(\mathbf{r}, \mathbf{c}_a) = \mathbf{r} \cdot \mathbf{c}_a / (\lVert\mathbf{r}\rVert\ \lVert\mathbf{c}_a\rVert)$; $\mathbf{c}_a$ = mean golden agent turn of archetype a	2, 3	Higher is closer to the target style
Gold similarity	$\cos(\mathbf{r}, \mathbf{h})$; $\mathbf{h}$ = embedding of the human reply	2	Higher is closer to the human agent
Slot F1	$F_1 = 2PR / (P + R)$ over slot values	3	0 to 1; higher is more accurate
Exemplar leakage rate	$\lvert E_r \cap (E_x \setminus E_d)\rvert / \lvert E_r\rvert$; entities in response, exemplars, dialogue	3	0 to 1; lower is better
Rule adherence rate	$1 - V_t / A_t$; violating and all agent turns	3	0 to 1; higher is better

Metric	Formula	RQ	Interpretation
Turns to goal	Turns until all required slots are filled and an offer is accepted (log scale)	4	Lower is more economical
Redundant elicitation rate	R_{red} / A_t; requests for an already-informed slot over agent turns	4	0 to 1; lower is better

Note. Cosine measures use de-lexicalized all-MiniLM-L6-v2 embeddings. ARI follows Hubert and Arabie (1985). BERTScore (Zhang et al., 2020) is planned as a token-level check for the final report.

Preliminary Results

RQ1: Clustering Algorithms and Primary Archetypes (Complete)

On the full golden partition, Silhouette was highest at $k = 2$ (.206) and nearly flat from $k = 3$ to 5 (.141 to .148), with no clear Davies–Bouldin or inertia elbow, so the pre-specified $k = 3$ was retained to give interpretable archetypes with enough exemplars (smallest cluster 24.1%; see Table E4 and Figure E1 in Appendix E for the diagnostics).

Table 9 compares the algorithms. HDBSCAN produced the highest mean Silhouette (M = .211, SD = .044), followed by K-Means (M = .147, SD = .019) and agglomerative clustering (M = .118, SD = .040), and all three far exceeded the permutation baseline ($M = -.013$). The one-way ANOVA was significant, $F(2, 156) = 91.33$, $p < .001$, $\eta^2 = .54$, and held under Welch's correction, $F(2, 89.15) = 66.98$, $p < .001$, and the Kruskal–Wallis test, $H(2) = 86.82$, $p < .001$. Every pairwise difference was significant under Tukey and Games–Howell comparisons (Table E5; runs plotted in Figure E2). The null hypothesis is rejected.

Two qualifications matter. HDBSCAN's advantage depends on its settings: with the common minimum cluster size of 15 it found no clusters in 53 of 53 subsamples, and with a minimum of 5 it still labeled 39.1% of dialogues as noise and found 2 to 6 clusters (most often 2), so its higher Silhouette partly reflects fewer clusters. All Silhouette values are below .25,

which indicates weak structure: archetypes are tendencies on a continuum. Because routing needs a stable number of archetypes and a centroid for every dialogue, K-Means with $k = 3$ defined the archetypes.

Table 9

Cluster Quality by Algorithm

Algorithm	*n* runs	Silhouette *M*	Silhouette *SD*	Davies–Bouldin *M*	Davies–Bouldin *SD*
K-Means	53	.147	.019	3.129	0.113
Agglomerative (Ward)	53	.118	.040	3.550	0.269
HDBSCAN	53	.211	.044	2.657	0.241
Permutation baseline	53	−.013	.006	—	—

Note. Each run clusters an independent random subsample of 500 golden dialogues. The permutation baseline shuffles K-Means labels within each subsample.

Table 10 profiles the archetypes on features not used for clustering; names follow a fixed rule (most revisions = Constraint Negotiator; longer of the other two = Guided Explorer). Direct Transactional dialogues are short (9.9 turns) with almost no offers (0.84). Constraint Negotiators write the longest utterances and revise slot values most often (20.3). Guided Explorers form the largest group (50.2%) and have the longest dialogues (17.2 turns) but the shortest user utterances, indicating an agent-led search. Task success is high in every archetype, so they differ in how users converse rather than in whether they succeed.

Archetype was associated with the Frames user role-player, $\chi^2(20, N = 1{,}334) = 92.63$, $p < .001$, Cramér's $V = .19$, so individual role-player style contributes to the archetypes. The frozen centroids generalized within Frames: centroid similarity did not differ between golden and held-out dialogues, $t(754.9) = 0.42$, $p = .676$, and held-out proportions matched the golden ones, $\chi^2(2, N = 401) = 4.69$, $p = .096$. They did not transfer to MultiWOZ: MultiWOZ dialogues were far

less similar to their nearest Frames centroid (M = .805 vs. .894), $t(576.2) = 38.29$, $p < .001$, and 84.0% fell into one archetype (Table E6 and the UMAP projection in Figure E3), which supports keeping MultiWOZ descriptive.

Table 10

Behavioral Profiles of the Three Primary Archetypes (Golden Partition)

Feature	Direct Transactional	Constraint Negotiator	Guided Explorer
n (%)	240 (25.7)	225 (24.1)	468 (50.2)
Turns, *M* (*SD*)	9.92 (5.02)	15.64 (6.84)	17.19 (6.10)
Words per user utterance	10.60	12.94	9.63
Words per agent utterance	11.67	18.15	15.71
Slot-value revisions	6.92	20.32	16.68
Hesitation markers	0.67	1.15	1.15
Agent offers	0.84	4.28	3.73
Budget stated (%)	63.7	70.7	75.4
Task success (%)	92.1	94.2	95.3

Note. Values are means per dialogue unless stated. Revisions count changes of a slot's value across user and agent acts.

Interim answer. The algorithms differ significantly with a large effect. HDBSCAN separates best but only with relaxed settings, heavy noise, and an unstable number of clusters, whereas K-Means gives a stable three-archetype solution that generalizes to held-out Frames dialogues but not to MultiWOZ. The pre-specified answer to RQ1 is the three primary archetypes, which were used in RQ2.

RQ1 Sensitivity Analysis: Refined Archetypes (Complete)

Three diagnostics showed that the primary archetypes are a weak basis for routing: a router predicting the archetype from the first five turns reached only $\kappa = .11$ on held-out dialogues, the trajectory vectors mix user and agent behavior, and the archetypes are partly tied to role-players. With Gemini embeddings, the 159-run algorithm comparison reproduced the

primary result with larger effects (η^2 = .50 for turn vectors and .62 for document vectors, both p < .001; Table D3), so the answer to RQ1 does not depend on the embedding model. Restricting MiniLM vectors to user turns raised three-cluster stability from ARI = .72 to .87 and lowered the role-player association from V = .19 to .12.

On whole conversations the hybrid representation was unstable (best eligible solution k = 4, ARI = .67) and largely reproduced how the dialogue ended (three-cluster association with the rated outcome V = .57), because late messages record the result of the search rather than the user's approach. Table 11 shows that truncating to the first M user messages solved this: three-cluster solutions were stable at every cut-off (ARI = .84 to .89), no solution with four or more clusters exceeded ARI = .69, and the outcome association rose with more messages (V = .16 at three to .45 at nine; all panels in Figure E4). The pre-declared rule selected the first nine user messages with k = 3 (ARI = .89; 93% core members that keep their archetype in at least 90% of 30 refits; smallest archetype 26%).

Table 11

Stability, Routing, and Outcome Association of Hybrid Archetypes by Conversation Length

User messages used	Dialogues shortened (%)	Stability ARI	Core members (%)	Smallest cluster (%)	Routing κ	Outcome V	Best ARI, $k \geq 4$
3	94.7	.89	92.7	18.3	.72	.16	.67
5	78.2	.84	77.1	20.3	.38	.27	.63
7	53.2	.85	89.9	29.7	.25	.38	.69
9	30.7	.89	93.1	25.5	.24	.45	.63
Whole conversation	0.0	.62	53.7	7.9	.22	.57	.67

Note. Values are for k = 3 on the golden partition (N = 933) unless stated. Stability is the mean ARI between K-Means fits on 50 pairs of random 80% subsamples; core members keep their archetype in at least 90% of 30 refits. Routing κ compares the archetype of held-out dialogues

with a logistic-regression prediction from their first five turns. Outcome *V* is Cramér's *V* with the LLM-rated outcome of the whole conversation. The adopted solution uses nine messages. ARI = adjusted Rand index.

Table 12 profiles the refined archetypes; names follow a fixed rule (highest destination flexibility = Open Explorer; more decisive of the other two = Decisive Booker). Decisive Bookers (45.0%) are polite and decisive, pursue one specific trip (99%), and book in 90% of dialogues. Open Explorers (25.5%) are flexible about the destination, consider many alternatives, revise often, and have the longest dialogues (20.8 turns). Rigid-Constraint Seekers (29.5%) have a fixed destination and dates and ask few questions; their dialogues are short (10.6 turns) and end without a suitable option in 68% of cases. Frames task success does not distinguish them (Cramér's *V* = .08).

Table 12

Profiles of the Three Refined Archetypes (Golden Partition)

Feature	Decisive Booker	Open Explorer	Rigid-Constraint Seeker
n (%)	420 (45.0)	238 (25.5)	275 (29.5)
Turns, *M* (*SD*)	14.49 (4.54)	20.78 (6.95)	10.60 (5.73)
User messages, *M*	7.58	10.73	5.55
Destination flexibility (1–5)	2.54	3.49	2.20
Date flexibility (1–5)	2.57	2.77	2.18
Exploration breadth (1–5)	1.97	2.96	1.59
Decisiveness (1–5)	3.97	2.58	3.23
Detail orientation (1–5)	2.39	2.87	1.46
Revision tendency (1–5)	2.00	3.03	2.40
Question asking (1–5)	2.05	2.72	1.43
Politeness (1–5)	4.26	3.80	3.07
Price sensitivity (1–5)	3.20	3.18	3.12
Information given up front (1–5)	2.18	1.82	2.16
Goal: explore destinations (%)	0.5	39.1	2.2
Goal: book a specific trip (%)	99.3	50.8	89.8

Feature	Decisive Booker	Open Explorer	Rigid-Constraint Seeker
Outcome: booked (%)	90.5	68.1	28.0
Outcome: no suitable option (%)	6.2	18.5	68.0
Frames task success (%)	94.5	96.2	92.0

Note. Ratings and goals are LLM annotations of the first nine user messages (the clustering input); outcomes are LLM annotations of the whole conversation and were not used for clustering.

The refined and primary archetypes overlap only partly (ARI = .20). On held-out dialogues the refined archetypes reproduced the golden proportions, $\chi^2(2, N = 401) = 0.93$, $p = .628$, and routing after five turns improved from $\kappa = .11$ to .24, which is still only fair. An ablation (Table E7) showed that truncation is the main driver of the gain: with the behavior profile, stability rose from ARI = .56 to .73 on whole conversations to .85 to .87 on nine messages; alternative representations agree only partly with the adopted labels (ARI = .14 to .45).

Decision. The primary archetypes remain the pre-specified answer to RQ1 and were used in RQ2. RQ3 and RQ4 use the refined archetypes because they are more stable, describe the user rather than the whole dialogue, and are recognized earlier; RQ2 will be repeated with them. They remain regions of a continuum (Silhouette only .033 above its permutation baseline) and partly capture whether a suitable package was available (outcome $V = .45$).

RQ2: Cluster-Adaptive Few-Shot Prompting (Complete)

The pilot ran every step except the LLM call on the experimental sample (Tables E8 and E9; Figure E5). Routing from a three-turn prefix was correct for 12 of 33 dialogues (36.4%), close to the 33.3% expected by chance. Exemplar relevance differed by strategy, $F(1.11, 35.41) = 386.65$, $p < .001$, partial $\eta^2 = .92$: both similarity-based strategies retrieved far more relevant

exemplars than the static set, and global retrieval was slightly more relevant than cluster-adaptive retrieval. The style metric was valid but modest: human replies were closer to their own archetype's centroid than to the others, $t(32) = 2.60$, $p = .014$, $d_z = .45$.

Table 13 shows the experimental results. Style convergence did not differ by condition, $F(1.90, 60.95) = 1.08$, $p = .342$, partial $\eta^2 = .03$, and the Friedman test agreed, $\chi^2(2, N = 33) = 5.72$, $p = .057$. Zero-shot responses were descriptively closest to the target centroid ($M = .519$), the opposite of the hypothesized order, no Holm-adjusted contrast was significant (Table E10; Figure E6), and similarity to the human reply did not differ either. The null hypothesis for RQ2 is not rejected.

The exploratory length measure showed a large effect: zero-shot responses averaged 25.2 words, whereas static and cluster-adaptive responses averaged 10.6 and 12.3 words, close to the human replies ($M = 10.5$), partial $\eta^2 = .52$; the two few-shot conditions did not differ from each other (Holm $p = .196$).

Table 13

Generated Responses by Prompting Condition

Outcome	Zero-shot *M* (*SD*)	Static few-shot *M* (*SD*)	Cluster-adaptive *M* (*SD*)	*F*	*p*	η_p^2
Style convergence (cosine)	.519 (.126)	.491 (.109)	.486 (.099)	1.08	.342	.03
Similarity to human reply (cosine)	.317 (.195)	.325 (.223)	.333 (.221)	0.14	.861	.00
Response length (words)	25.2 (11.8)	10.6 (7.0)	12.3 (8.2)	34.36	< .001	.52

Note. $N = 33$ held-out dialogues, each measured under all three conditions; gemini-2.5-flash. *p* values use the Greenhouse–Geisser correction. Response length is exploratory.

Interim answer. Any human examples move the model toward human pacing, but archetype-matched examples add nothing measurable beyond fixed examples. Two likely

reasons are near-chance routing, so most dialogues received exemplars from the wrong archetype, and weak archetype separation with a modest style metric; the observed effect (f = 0.18) is below the medium effect assumed in the power analysis.

RQ3: Rules Versus Prompting (Groundwork Complete)

The rule engine passed all 14 boundary tests. The corpus audit (Table E11) shows that only two rules can be checked from Frames annotations: a user-stated number of adults appears in 24.1% of dialogues and triggers the capacity rule in 54, and a numeric budget appears in 65.1% and triggers the minimum-budget rule in 266. The visa, passport, and discount rules can never be triggered by public data, which is the empirical basis for injecting synthetic scenarios. RQ3 is not yet answered.

RQ4: Turn Economy Across Archetypes (Groundwork Complete)

Table 14 gives the human baseline for all 1,334 Frames dialogues by refined archetype. Turn counts differed by archetype, $H(2) = 433.22$, $p < .001$, as expected by construction (distributions in Figure E7). Redundant elicitation was rare among human agents ($M = .015$, about one request in 66 agent turns) and did not differ by archetype, $H(2) = 4.12$, $p = .128$. The synopsis expectation of reducing RER from .16 is therefore not supported; RER will instead be compared between the two LLM agents.

Table 14

Human Baseline of Turn Economy by Refined Archetype (Frames)

Archetype	*n*	Turns *M* (*SD*)	RER *M* (*SD*)	Agent request turns *M*
Decisive Booker	595	14.56 (4.52)	.014 (.047)	1.44
Open Explorer	337	20.75 (7.07)	.016 (.049)	1.82
Rigid-Constraint Seeker	402	10.60 (5.59)	.017 (.061)	1.27

Note. RER = redundant elicitation rate. Archetypes for held-out dialogues were assigned with the K-Means model fitted on the golden partition.

Interim Limitations and Risks

Archetypes are weakly separated (primary Silhouette < .25; refined only .033 above the permutation baseline) and routing after five turns reaches only $\kappa = .24$, so routing errors dilute any effect of archetype-matched prompting. The primary archetypes are associated with role-players ($V = .19$) and the refined ones with the final outcome ($V = .45$), and other reasonable representations give partly different archetypes. The behavior ratings come from an LLM and were validated on whole conversations, RQ3 scenarios are synthetic, and RQ4 relies on a simulated user. Frames is role-play with fictional cities, so conclusions apply to vacation-package negotiation dialogues of this kind. The archetype definition was refined after the primary analysis (mitigated by a pre-declared rule and by reporting every configuration), and the observed RQ2 effect ($f = .18$) was smaller than the medium effect assumed for power.

Project risks are API quota limits (mitigated by caching every response), weak routing (refined archetypes and a global nearest-neighbor condition), an unrealistic simulator (persona check), and schedule slippage (analysis ends in Week 8, leaving Weeks 9 and 10 as buffer).

Next Steps

- Week 6: repeat RQ2 with the refined archetypes and global retrieval; prepare RQ3 scenarios.
- Week 7: run the RQ3 experiment (slot F1, exemplar leakage, rule adherence).
- Week 8: RQ4 simulation, persona and annotation-reliability checks, BERTScore, and robustness checks.
- Weeks 9 and 10: final presentation, final report, and repository release (Appendix B).

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Appendix A

Repository Structure and Replication Guide

All materials are available at https://github.com/surnella/QM640-Dialogue-Clustering-Capstone. Table A1 lists the parts of the repository used in this report, and Table A2 lists the commands that reproduce every table and figure.

Table A1

Repository Structure

Path	Contents
data/raw/	Downloaded Frames and MultiWOZ 2.2 corpora
data/canonical/	Canonical two-tier dialogue file
data/interim/	Cleaned samples, embeddings, archetypes, LLM annotations, prompts, and run summaries
data/embeddings/gemini/	Cached Gemini embeddings (regenerated by step 2c; not versioned)
src/ingestion.py, src/canonical_schema.py	Corpus parsers and Pydantic schema
src/rule_engine.py, rules/	Declarative rule engine and YAML rule file
src/interim_pipeline/	Reproducible analysis steps 0 to 4, including steps 2b to 2i (this report)
src/report/, src/build_interim_submission.py	APA report builder
outputs/interim/tables/, outputs/interim/figures/	Result tables (CSV) and figures (PNG)
requirements.txt	Python dependencies

Table A2

Replication Commands

Step	Command (run from src/interim_pipeline/)	Produces
0	python step0_sample_size.py	Table 6
1	python step1_prepare_data.py	Tables 4, 5, and E3; Figures 1 and 2
2	python step2_rq1_clustering.py	Tables 9, 10, and E4 to E6; Figures E1 to E3
2b	python step2b_rq1_representation.py --emb minilm	User-turn and dialogue-act representations
2c	python step2c_gemini_embeddings.py	Gemini embeddings (requires GEMINI_API_KEY)
2d	python step2d_rq1_gemini_algorithms.py	Table D3
2e	python step2e_llm_user_annotation.py (also --retest and --max-user-msgs 3, 5, 7, 9)	LLM behavior annotations (requires GEMINI_API_KEY)
2f	python step2f_rq1_llm_hybrid.py	Table D2; whole-conversation hybrid
2g	python step2g_rq1_truncation.py	Table 11; Figure E4; refined labels
2h	python step2h_rq1_ablation.py	Table E7
2i	python step2i_rq1_refined_archetypes.py	Tables 12 and 14; Figure E7
3	python step3_rq2_pipeline.py	Tables E8 and E9; Figure E5; saved prompts
3 (execution)	python step3_rq2_pipeline.py --execute	Tables 13 and E10; Figure E6 (requires GEMINI_API_KEY)
4	python step4_rq3_rq4_groundwork.py	Tables 7 and E11
Report	python src/build_interim_submission.py (from the repository root)	This document

Note. Python 3.11; all random steps use seed 42. LLM and embedding responses are cached, so steps 2c and 2e only call the API for missing items.

Appendix B

Ten-Week Project Schedule

Table B1 maps the project to the 10-week QM640 schedule. All analysis finishes in Week 8, and Weeks 9 and 10 are reserved for the presentation and final report.

Table B1

Ten-Week Schedule Aligned With the QM640 Course Topics

Week (dates)	Course topic	Project tasks and deliverables	Status
1 (Sep 27–Oct 3)	Define research problem and topic	Problem statement finalized; repository set up; Frames and MultiWOZ downloaded	Complete
2 (Oct 4–10)	Synopsis draft	Research questions and two-tier schema; synopsis draft (Oct 6)	Complete
3 (Oct 11–17)	Literature search and data collection	Synopsis final (Oct 13); ingestion, canonicalization, and embeddings	Complete
4 (Oct 18–24)	Objectives and hypotheses	Quality audit, EDA, power analysis, RQ1 clustering; RQ2 pilot; RQ3 and RQ4 groundwork; interim draft (Oct 20)	Complete
5 (Oct 25–31)	Research design and methodology	Interim report final (Oct 27); RQ2 generation run and analysis (complete); RQ1 sensitivity analysis of the archetype definition (complete)	In progress
6 (Nov 1–7)	Data analysis plan	RQ2 repeated with refined archetypes and global k-NN; RQ3 scenario injection; final report draft (Nov 3)	Planned
7 (Nov 8–14)	Data analysis plan execution	RQ3 2 × 2 experiment and analysis	Planned
8 (Nov 15–21)	Data analysis assessment	RQ4 simulation, 2 × 3 analysis, robustness checks; analysis complete	Planned
9 (Nov 22–28)	Refinement and documentation	Final presentation (Nov 27–28)	Planned
10 (Nov 29–Dec 5)	Document results and findings	Final report and repository release	Planned

Note. EDA = exploratory data analysis. No tasks are scheduled outside Weeks 1 to 10.

Appendix C

Comparison With Related Research

Table C1 compares the studies closest to this capstone and shows how the present design extends them.

Table C1

Related Research and Its Relationship to the Present Study

Study	Main contribution	Connection to this study	Gap addressed here
El Asri et al. (2017), Frames	1,369 human–human vacation-package dialogues with frame tracking	Primary corpus for embeddings, archetypes, and exemplars	Does not cluster dialogues or use them for prompting
Zang et al. (2020), MultiWOZ 2.2	Corrected multi-domain dialogue states and schema	External projection check and schema mapping	Focuses on slot tracking, not archetypes or prompting
Liu et al. (2022), KATE	Similarity-based in-context example selection beats random selection	Motivates retrieval; global nearest neighbor is a comparison	Single-turn tasks; no style or leakage measures
Rebedea et al. (2023), NeMo Guardrails	Programmable rails outside the LLM	Motivates the decoupled rule engine	Does not adapt style or test interaction with prompting
Yao et al. (2024), τ-bench	Agent evaluation against explicit domain policies	Supports measuring rule adherence separately	Tool-calling agents; no archetype adaptation
Acikgoz et al. (2025), TD-EVAL	Turn-level and dialogue-level evaluation	Supports combining similarity with turn-level checks	Evaluation framework only; no adaptive prompting

Note. The comparison describes relevance to this study and does not imply that the cited studies used the same data or metrics.

Appendix D

Behavior Annotation Codebook and Quality Checks

Table D1 gives the codebook used by the LLM to rate each Frames user in the RQ1 sensitivity analysis. The model read the conversation (whole or truncated after the *M*-th user message) and rated only the user, using the agent's messages as context. Table D2 reports the reliability and criterion validity of the ratings, and Table D3 repeats the RQ1 algorithm comparison with Gemini embeddings.

Table D1

LLM Behavior Annotation Codebook

Field	Type	Definition and scale anchors
Primary goal	Category	Main purpose: explore destinations, price shopping, book a specific trip, compare options, information only, or other
Secondary goals	Categories	Other purposes clearly present (same categories; may be empty)
Price sensitivity	1–5	1 = price never matters; 5 = price dominates every decision
Destination flexibility	1–5	1 = one fixed destination; 5 = open to anywhere
Date flexibility	1–5	1 = fixed dates; 5 = fully flexible dates
Exploration breadth	1–5	1 = considers a single option; 5 = considers many alternatives
Decisiveness	1–5	1 = hesitant, cannot decide; 5 = decides quickly and firmly
Detail orientation	1–5	1 = no interest in specifics; 5 = asks about many details
Information up front	1–5	1 = reveals requirements bit by bit; 5 = states all requirements in the first message
Revision tendency	1–5	1 = never changes requirements; 5 = changes them repeatedly
Question asking	1–5	1 = asks no questions; 5 = asks many questions
Politeness	1–5	1 = curt or rude; 5 = very polite
Emotional tone	Category	Positive, neutral, impatient, or frustrated
Outcome	Category	Booked, no suitable option, undecided or left, or unclear (not used for clustering)
Party type	Category	Solo, couple, family, group, business, or unknown (not used for clustering)
Unusual conversation	Yes/no	Off-task, nonsensical, or unlike a normal travel request

Note. Gemini 2.5 Flash, temperature 0, internal reasoning disabled, output constrained to a JSON schema with these fields. The instruction asked the model to base every rating on evidence in the user's messages and to use 3 for moderate or mixed behavior.

Table D2

Reliability and Criterion Validity of the LLM Behavior Ratings

Field	Re-annotation κ	Behavior counted from Frames acts	Spearman ρ
Destination flexibility	.96	—	—
Date flexibility	.89	—	—
Exploration breadth	.95	Frame switches per user turn	.30
		Comparison requests per user turn	.26
Decisiveness	.91	—	—
Detail orientation	.89	Distinct preference slots mentioned	.45
Revision tendency	.89	Slot-value changes in user acts	.73
Question asking	.90	Request acts per user turn	.65
Politeness	.91	—	—
Price sensitivity	.95	Budget stated	.28
Information given up front	.87	Share of slots stated in the first user turn	.32
Primary goal	.78	—	—
Emotional tone	.88	—	—
Outcome	.98	—	—
Party type	.93	—	—

Note. Re-annotation: $n = 100$ random dialogues annotated a second time at temperature 0.7; quadratic-weighted κ for ratings and Cohen's κ for categories. Criterion validity: $N = 1{,}334$ dialogues; all ρ $p < .001$. Both checks use whole-conversation annotations.

Table D3

Algorithm Comparison Repeated With Gemini Embeddings

Representation	Algorithm	Silhouette *M* (*SD*)	ANOVA	η^2
Decay-weighted turn vectors	K-Means	.158 (.012)	$F(2, 156) = 77.09$	.50
	Agglomerative	.150 (.062)		
	HDBSCAN	.289 (.093)		
	Permutation baseline	−.015 (.009)		
Whole-dialogue document vectors	K-Means	.161 (.059)	$F(2, 156) = 127.47$	.62
	Agglomerative	.213 (.067)		
	HDBSCAN	.345 (.057)		
	Permutation baseline	−.019 (.012)		

Note. Each algorithm was run on 53 random subsamples of 500 golden dialogues, as in Table 9, using gemini-embedding-001 vectors (3,072 dimensions). All ANOVAs $p < .001$; Welch and Kruskal–Wallis tests agreed.

Appendix E

Supplementary Tables and Figures

This appendix holds second-level material that supports the conclusions stated in the body. Each table and figure is summarized where it is cited in the body.

Data Description

Table E1

Data Dictionary, Raw Source Fields Used

Corpus	Field	Type	Description	Use
Frames	id	String	Unique dialogue identifier	Key
Frames	user_id, wizard_id	String	Role-player identifiers (11 users, 12 wizards)	Confound check
Frames	turns[].author	Category	user or wizard	Speaker tag
Frames	turns[].text	String	Utterance text	Embedding
Frames	turns[].timestamp	Float	Collection time (Unix ms)	Not used
Frames	turns[].labels.acts[].name	Category	Dialogue act (e.g., inform, request, offer, switch_frame)	Features, RER
Frames	turns[].labels.acts[].args[].key/val	String	Slot name and value (e.g., dst_city, budget)	Tier 1 slots
Frames	labels.wizardSurveyTaskSuccessful	Boolean	Wizard-reported task success	Stratify, pool
Frames	labels.userSurveyRating	Float	User satisfaction rating (1–5)	Descriptive
MultiWOZ	dialogue_id	String	Unique dialogue identifier	Key
MultiWOZ	services	List	Services involved (hotel, train, taxi, ...)	Inclusion
MultiWOZ	turns[].speaker	Category	USER or SYSTEM	Speaker tag
MultiWOZ	turns[].utterance	String	Utterance text	Embedding
MultiWOZ	turns[].frames[].state.slot_values	Map	Cumulative dialogue state per service	Tier 1 slots

Note. RER = redundant elicitation rate. Only the fields listed were read; all other fields were ignored.

Table E2

Data Dictionary, Canonical Schema Fields

Field	Type	Description
dialogue_id, source_dataset	String	Identifier and corpus of origin (frames or multiwoz)
total_turns, user_turns, wizard_turns	Integer	Turn counts
turns[].turn_id, speaker, utterance	Mixed	Ordered turns with corpus-neutral speaker and text
turns[].acts, slots_mentioned, revisions	List, map	Acts, slots informed, and slots whose value changed in the turn
Tier 1: origin_city, destination_city	String	Trip endpoints
Tier 1: departure_date, return_date, duration_nights	String, integer	Trip timing
Tier 1: traveler_count_adults, traveler_count_children	Integer	Party size
Tier 1: budget_amount, accommodation_tier	Float, string	Budget ceiling and hotel tier
Tier 2: visa_processing_window_days, passport_expiry_months	Integer	Policy timing (synthetic in RQ3)
Tier 2: package_type, sharing_basis, discount_applied_percentage	String, float	Package, room, and discount (synthetic in RQ3)
revisions_count	Integer	Number of slot-value changes across the dialogue
hesitation_markers_count	Integer	User hesitation phrases (e.g., "maybe," "not sure")
budget_specified	Boolean	Whether a numeric budget was stated

Note. Schema implemented as Pydantic models in src/canonical_schema.py.

Table E3

Observed Slot Completeness by Corpus

Slot	Tier	Frames (%)	MultiWOZ 2.2 (%)
origin city	Tier 1	99.4	70.8
destination city	Tier 1	99.0	71.4
departure date	Tier 1	84.9	68.6
return date	Tier 1	70.8	0.0
duration nights	Tier 1	61.8	27.0
traveler count adults	Tier 1	63.9	54.0
traveler count children	Tier 1	18.4	0.0
budget amount	Tier 1	72.9	24.1
accommodation tier	Tier 1	72.7	19.7
visa processing window days	Tier 2	0.0	0.0
passport expiry months	Tier 2	0.0	0.0
discount applied percentage	Tier 2	0.0	0.0

Note. Percentage of dialogues in which the slot is mentioned at least once in the dialogue-act (Frames) or dialogue-state (MultiWOZ) annotations. MultiWOZ budget is the categorical price range. Frames N = 1,334; MultiWOZ N = 1,499. Figure 2 in the body plots these values.

RQ1: Clustering Diagnostics

Table E4

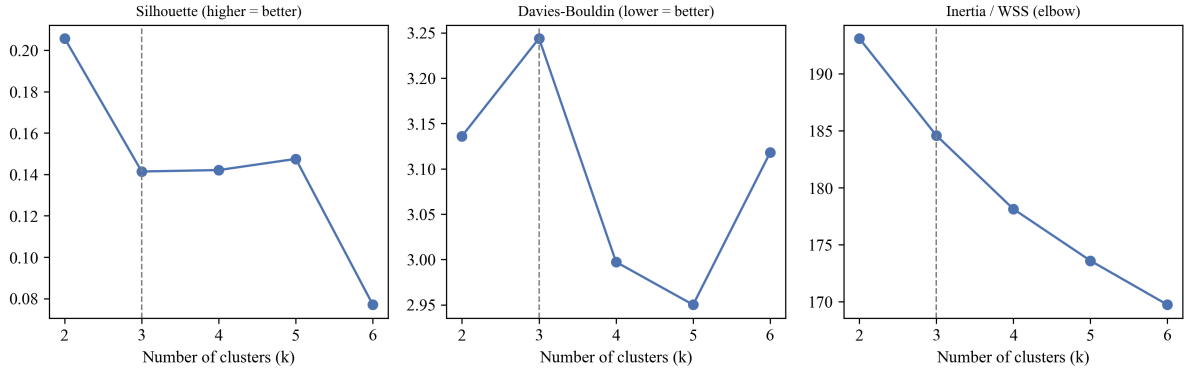
Choice of the Number of Clusters (K-Means on the Golden Partition)

k	Silhouette	Davies–Bouldin	Inertia	Smallest cluster (%)
2	.206	3.136	193.1	35.5
3	.141	3.244	184.6	24.1
4	.142	2.997	178.1	17.7
5	.148	2.950	173.6	8.7
6	.077	3.118	169.7	8.4

Note. N = 933 golden dialogues; cosine Silhouette.

Figure E1

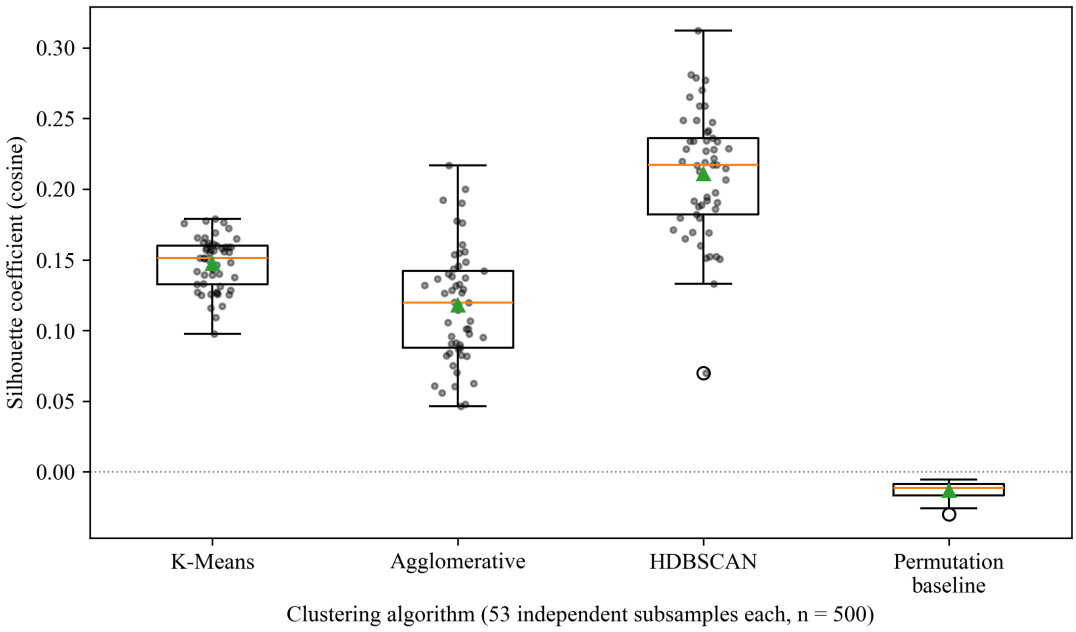
Cluster-Number Diagnostics



Note. The dashed line marks the pre-specified $k = 3$.

Figure E2

Silhouette Coefficients by Algorithm



Note. Each point is one subsample run; triangles mark means. The permutation baseline shows the Silhouette expected with no structure.

Table E5

Pairwise Comparisons of Mean Silhouette

Comparison	Mean difference	95% CI (Tukey)	*p* (Tukey)	*p* (Games–Howell)
K-Means − Agglomerative	.029	[.013, .046]	< .001	< .001
HDBSCAN − Agglomerative	.093	[.076, .110]	< .001	< .001
HDBSCAN − K-Means	.063	[.047, .080]	< .001	< .001

Note. CI = confidence interval. Games–Howell comparisons do not assume equal variances.

Table E6

Archetype Assignment and Centroid Similarity Across Samples

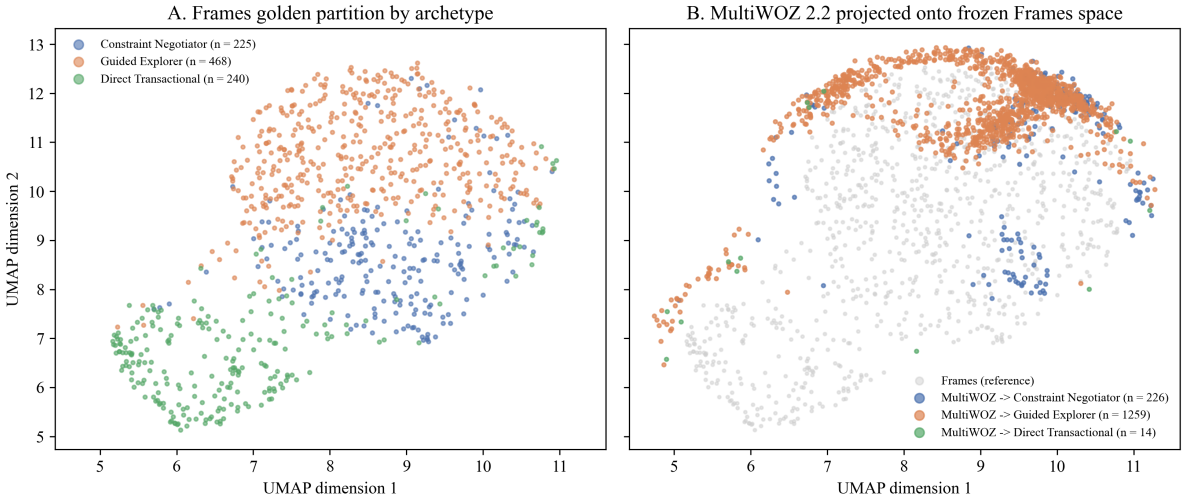
Sample	*N*	Centroid similarity *M* (*SD*)	Direct Transactional (%)	Constraint Negotiator (%)	Guided Explorer (%)
Frames golden	933	.895 (.042)	25.7	24.1	50.2
Frames held-out	401	.894 (.042)	29.4	20.2	50.4
MultiWOZ 2.2	1,499	.805 (.037)	0.9	15.1	84.0

Note. Centroid similarity is the cosine between a dialogue vector and its nearest frozen centroid.

Centroids were fitted on the golden partition only.

Figure E3

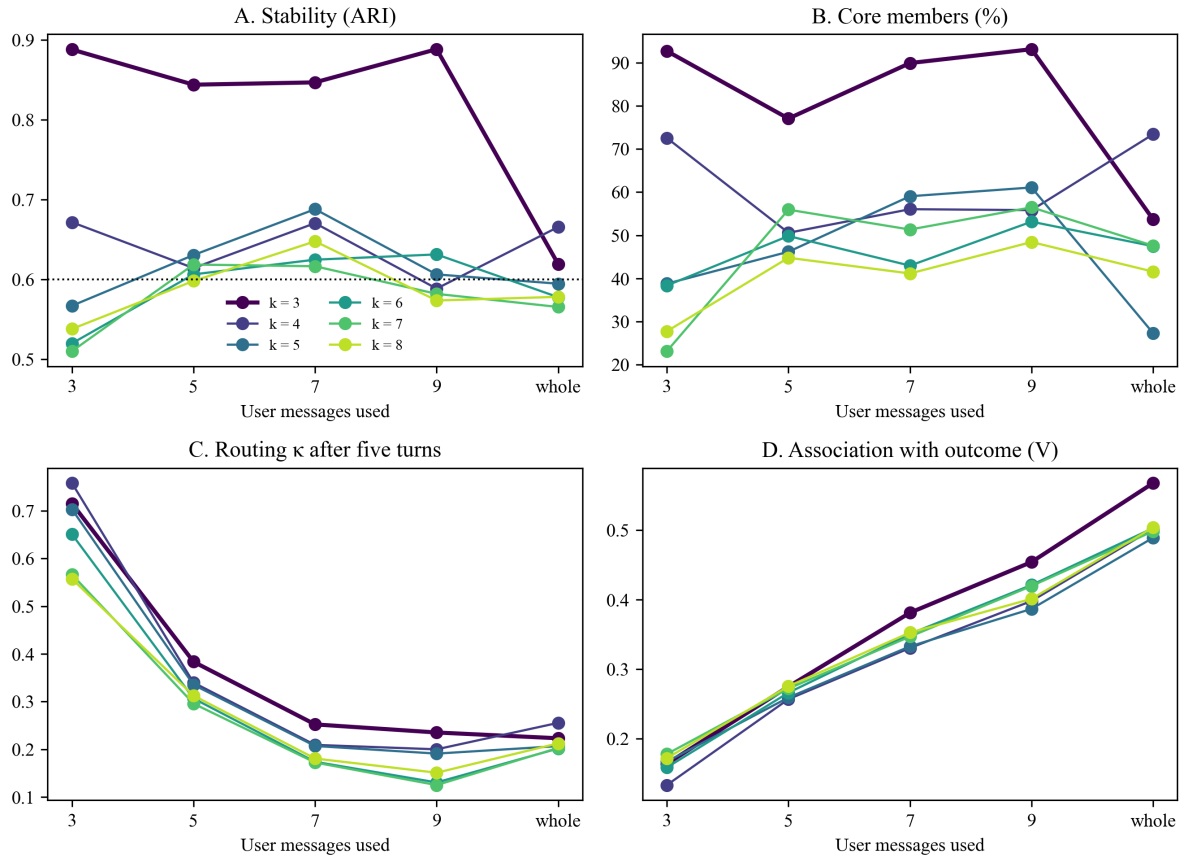
Archetypes in the Frames Embedding Space and Projection of MultiWOZ 2.2



Note. Two-dimensional UMAP projection (McInnes et al., 2018) fitted on the golden partition; MultiWOZ dialogues were transformed into the same projection without refitting.

Figure E4

Hybrid Archetype Diagnostics by Conversation Length and Number of Clusters



Note. Each line is one number of clusters (*k*). The dotted line in Panel A marks the stability threshold of .60. Panel C shows routing κ after five turns; Panel D shows Cramér's *V* between archetype and the rated outcome of the whole conversation.

Table E7

Ablation of the Refined Archetype Definition (Three Clusters)

Representation	Stability ARI (whole)	Stability ARI (9 messages)	Core members (%)	Outcome *V*	Routing κ	ARI with adopted labels
Gemini embedding + Gemini behavior profile (adopted)	.56	.87	89.2	.45	.24	1.00
MiniLM embedding + Gemini behavior profile	.69	.85	87.7	.40	.21	.33
Gemini behavior profile only	.73	.85	97.6	.35	.26	.17
Gemini embedding only	.91	.91	92.0	.34	.30	.45
MiniLM embedding only	.68	.72	72.1	.27	.32	.14
MiniLM embedding + dialogue-act features	.61	.76	88.0	.45	.25	.16

Note. Core members, outcome *V*, routing κ, and agreement with the adopted labels refer to the nine-message version. Behavior profile = LLM ratings with goal and tone indicators; dialogue-act features are counted from the Frames annotations without an LLM. Bootstrap stability varies by about .03 between runs, which explains small differences from Table 11. ARI = adjusted Rand index.

RQ2: Pilot and Experimental Details

Table E8

Pilot Exemplar Relevance by Retrieval Strategy

Strategy	Cosine to prefix *M*	*SD*	Exemplars from target archetype (%)
Static few-shot (fixed)	.578	.057	33.3
Global k-NN (unclustered)	.840	.054	36.4
Cluster-adaptive (routed sub-index)	.815	.060	36.4

Note. $N = 33$ held-out dialogue prefixes; three exemplars per prefix. Relevance is the mean cosine between the prefix and the retrieval keys of the selected exemplars.

Table E9

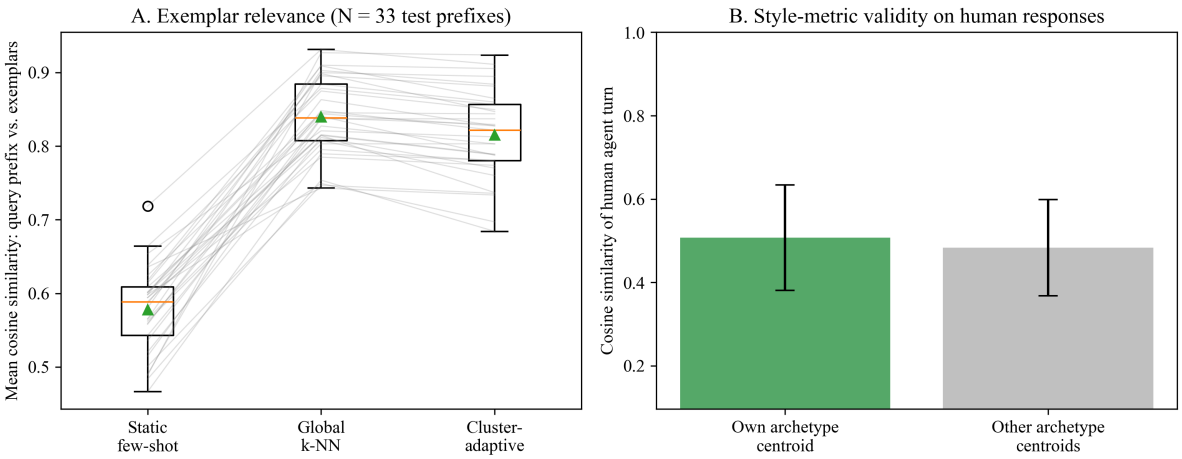
Holm-Adjusted Paired Contrasts of Exemplar Relevance

Contrast	Mean difference	*t*	*df*	*p* (Holm)	d_z
Global k-NN vs. Static	.262	22.00	32	< .001	3.83
Cluster-adaptive vs. Static	.237	18.18	32	< .001	3.16
Cluster-adaptive vs. Global k-NN	−.024	−6.93	32	< .001	−1.21

Note. Positive differences favor the first-named strategy.

Figure E5

RQ2 Pilot: Exemplar Relevance and Style-Metric Validity



Note. Panel A: lines connect the three strategies for the same prefix. Panel B: cosine similarity of the human agent reply to its own versus the other archetype centroids; error bars show 1 *SD*. The nearest centroid matched the archetype for 48.5% of replies.

Table E10

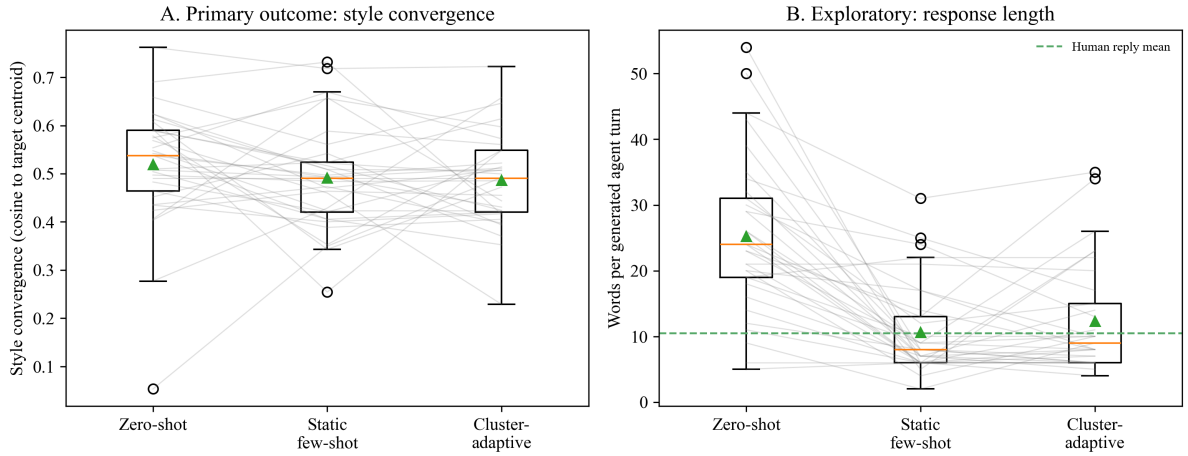
Holm-Adjusted Paired Contrasts for the RQ2 Outcomes

Outcome	Contrast	Mean difference	*t*	*df*	*p* (Holm)	d_z
Style convergence	Cluster-Adaptive vs. Zero-shot	−.033	−1.23	32	.678	−0.21
Style convergence	Static vs. Zero-shot	−.028	−1.20	32	.678	−0.21
Style convergence	Cluster-Adaptive vs. Static	−.005	−0.21	32	.836	−0.04
Similarity to human reply	Cluster-Adaptive vs. Zero-shot	.016	0.59	32	> .999	0.10
Similarity to human reply	Static vs. Zero-shot	.008	0.26	32	> .999	0.05
Similarity to human reply	Cluster-Adaptive vs. Static	.008	0.25	32	> .999	0.04
Response length	Static vs. Zero-shot	−14.6	−6.86	32	< .001	−1.19
Response length	Cluster-Adaptive vs. Zero-shot	−12.9	−5.83	32	< .001	−1.02
Response length	Cluster-Adaptive vs. Static	1.7	1.32	32	.196	0.23

Note. Positive differences favor the first-named condition.

Figure E6

RQ2 Results: Style Convergence and Response Length by Prompting Condition



Note. Lines connect the three conditions for the same dialogue; triangles mark means. The dashed line in Panel B is the mean length of the human agent replies.

RQ3 and RQ4: Groundwork Details

Table E11

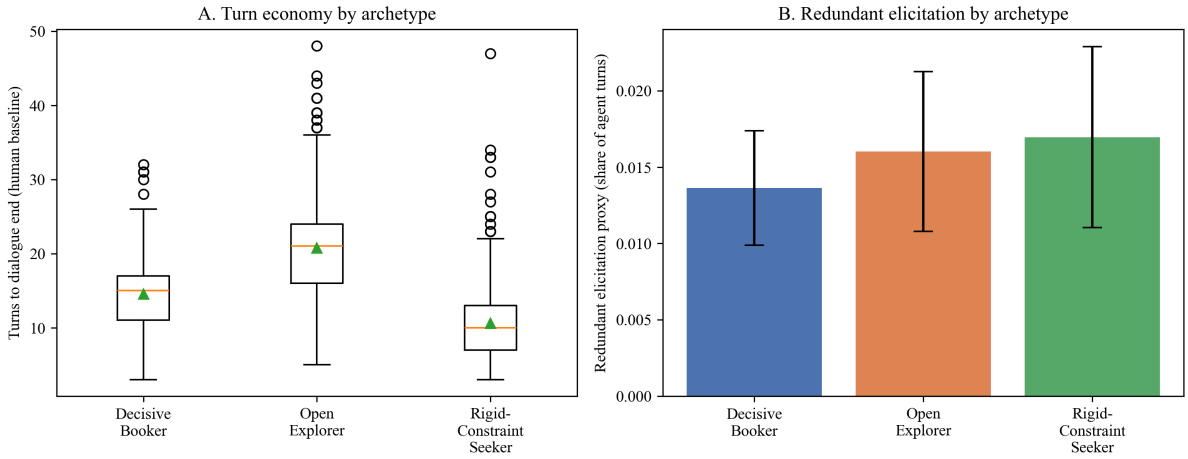
Corpus Audit of Rule Coverage in Frames

Rule	Required slot	Slot observed (%)	Dialogues triggering (*n*)	Triggering (%)
pax capacity	traveler count adults	24.1	54	4.05
budget minimum	budget amount	65.1	266	19.94
visa window	visa processing window days	0.0	0	0.00
passport validity	passport expiry months	0.0	0	0.00
discount threshold	discount applied percentage	0.0	0	0.00

Note. $N = 1{,}334$ Frames dialogues. Only numeric values stated by the user were counted, so the adult and budget percentages are lower than the any-mention rates in Table E3.

Figure E7

Human Baseline of Turn Economy and Redundant Elicitation by Refined Archetype



Note. Panel A: triangles mark means. Panel B: error bars show 95% confidence intervals.